

MODULE - 4

Mockito exercises

Exercise 1: Mocking and stubbing

The screenshot shows an IDE with a project structure on the left and a code editor on the right. The project structure includes a `main` directory and a `test` directory containing a `java` package with `org.example` sub-package. The `org.example` package contains several classes: `AppTest`, `AssertionsTest`, `Calculator`, `CalculatorAAATest`, `CalculatorTest`, `ExternalApi`, `MyService`, and `MyServiceTest`. The `MyServiceTest` class is selected in the project structure.

The code editor shows the `MyServiceTest` class with the following code:

```
8 public class MyServiceTest {
11     public void testExternalApi() {
12
13
14         ExternalApi mockApi = mock(ExternalApi.class);
15
16         when(mockApi.getData()).thenReturn(value: "Mock Data");
17
18         MyService service = new MyService(mockApi);
19
20         String result = service.fetchData();
21
22         assertEquals(expected: "Mock Data", result);
23     }
24 }
```

The Run tab at the bottom shows the test execution results:

```
Run MyServiceTest.testExternalApi x
MyServiceTest (org.exar 1 sec 358 ms) ✓ 1 test passed 1 test total, 1 sec 358 ms
testExternalApi 1 sec 358 ms
```

The console output shows the following warning:

```
WARNING: A Java agent has been loaded dynamically (C:\Users\BHASHINI P\.m2\repository\net\bytebuddy\
```

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Exercise 2: Verifying Interactions

The screenshot displays an IDE interface with the following components:

- Project Explorer:** Shows a project structure with `main` and `test` directories. Under `test`, there is a package `org.example` containing several test classes, including `VerifyInteractionTest`.
- Code Editor:** Displays the implementation of `VerifyInteractionTest.java`. The code includes package declarations, imports for JUnit and Mockito, and a test method `testVerifyInteraction()` that uses Mockito's `mock` and `verify` methods to verify an interaction with `ExternalApi`.
- Run Console:** Shows the execution of the test. The output indicates that the test passed successfully, with a message: `1 test passed 1 test total, 1 sec 358 ms`. Below this, there are several warning messages from the Java HotSpot(TM) 64-Bit Server VM regarding dynamic agent loading.

```
1 package org.example;
2
3 import org.junit.Test;
4
5 import static org.mockito.Mockito.*;
6
7 public class VerifyInteractionTest {
8
9     @Test
10     public void testVerifyInteraction() {
11
12         ExternalApi mockApi = mock(ExternalApi.class);
13
14         MyService service = new MyService(mockApi);
15
16         service.fetchData();
17
18         verify(mockApi).getData();
19     }
20 }
```

Run Console Output:

```
VerifyInteractionTest.testVerifyInteraction
1 test passed 1 test total, 1 sec 358 ms
testVerifyInteraction 1 sec 358 ms
WARNING: A Java agent has been loaded dynamically (C:\Users\BHASHINI P\m2\repository\net\bytebuddy\byte-buddy-agent\1.14.15\byte-buddy-agent-1.14.15.jar)
WARNING: If a serviceability tool is in use, please run with -XX:+EnableDynamicAgentLoading to hide this warning
WARNING: If a serviceability tool is not in use, please run with -Djdk.instrument.traceUsage for more information
WARNING: Dynamic loading of agents will be disallowed by default in a future release
Java HotSpot(TM) 64-Bit Server VM warning: Sharing is only supported for boot loader classes because bootstrap classpath has been appended
```